Executive Summary

This project focuses on building a complete sentiment analysis pipeline capable of transforming raw text data—such as customer reviews or social media posts—into meaningful, actionable insights. The work integrates core natural language processing techniques with machine learning models to evaluate sentiment and communicate results in a clear business context.

The project begins with the selection of a public sentiment dataset and proceeds through a structured workflow: text preprocessing, feature engineering, model development, and performance evaluation. Multiple classification models are trained and compared using metrics such as accuracy, precision, recall, and confusion matrices to ensure a rigorous assessment of model effectiveness.

Throughout the process, the project emphasizes best practices in data science, including working efficiently with limited computational resources, documenting methodological decisions, iterating on model improvements, and visualizing key findings. The final deliverable demonstrates not only technical proficiency in NLP and machine learning, but also the ability to interpret results and translate them into insights that support real-world decision-making.